

Salar Shakibhamedan

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Machine Learning Engineer, AI Researcher, and Data Scientist with experience in developing, optimizing, and deploying machine learning and deep learning models across academic and industrial settings. Strong background in efficient AI, computer vision, speech recognition, generative AI, LLM-based systems, and data-driven model development. Experienced in building Python-based AI pipelines, working with real-world data, conducting experiments on GPU/HPC environments, and translating research ideas into practical machine learning solutions.

Education

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| 2021 – Present | Ph.D. , TU-Wien, Vienna
<i>“Efficient and Scalable Optimization of Deep Learning Models (EU-Funded Project)”</i> |
| 2015 – 2018 | M.Sc. , K. N. Toosi University of Technology, Tehran
<i>“Multimodal Blind Source Separation”.</i> |
| 2010 – 2015 | B.Sc. , K. N. Toosi University of Technology, Tehran
<i>“(First) Persian Musical Instruments Recognition Model”</i> |

Experience

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| July 2021 – Present | Ph.D. Researcher , TU Wien, Vienna <ul style="list-style-type: none">- Developed and evaluated scalable deep learning optimization approaches for efficient AI model deployment.- Conducted experiments on high-performance computing resources and GPU-based environments.- Built reproducible Python-based research pipelines for model training, evaluation, and performance analysis. |
| Jan 2025 – July 2025 | Visiting Research Fellow , University of California, Irvine <ul style="list-style-type: none">- Designed and implemented end-to-end LLM pipelines for healthcare data generation, including embedding workflows and retrieval-based (RAG) architectures.- Built domain-specific RAG systems integrating heterogeneous medical data sources (documents, APIs, structured tabular data — real and synthetic).- Developed LLM-powered agentic AI systems for generating healthcare datasets with integrated fairness and bias mitigation mechanisms.- Evaluated, monitored, and validated LLM outputs for robustness, bias, and quality, collaborating with interdisciplinary stakeholders to deploy AI into real research workflows. |
| May 2020 – Apr 2021 | Machine Learning Engineer-Data Scientist , PART AI, Tehran <ul style="list-style-type: none">- Developed and deployed ML/DL models across computer vision, speech recognition, and recommendation systems, including one of the first Persian automatic speech recognition (ASR) models.- Performed data preprocessing, exploratory analysis, and feature engineering to extract insights, guide model design, and support real-world deployment.- Delivered production-ready ML solutions used in commercial applications.- Developed an OCR system for the Persian language.- Built and maintained Python-based AI systems in production environments. |
| Jun 2016 – Apr 2017 | Front Office (FO) Engineer/Data Analyst , MTN, Tehran <ul style="list-style-type: none">- Monitored and resolved network issues; found root causes and ensured stability- Applied AI-based models for issue analysis; generated professional reports and visualizations for operational insights |

Experience (continued)

Feb 2016 – Jan 2019

Teaching & Research Assistant, K. N. Toosi University, Tehran

- Contributed to research projects in digital signal processing and machine learning
- Assisted in teaching Advanced Digital Signal Processing and Digital Image Processing courses

Technical Skills

AI/ML: PyTorch, TensorFlow, Computer Vision (Transformer-Based, Advanced CNNs, OpenCV), Model Optimization, Data Analysis, AI System Design, Vision language models, Data Visualization, NLP

LLMs & Generative AI: End-to-end LLM workflow design, RAG architectures, embedding models, vector databases, prompt engineering, agentic workflows (LangChain, LangGraph), fine-tuning, OpenAI APIs, Microsoft Azure (incl. Copilot Studio), AWS/GCP, n8n.

Programming Languages: Python, C/C++, MATLAB, Java, R, SQL

DevOps & Tools: Git, Docker, Linux, LaTeX, Google Cloud

Complementary Skills

Miscellaneous: Academic research, teaching, training, and publishing

Languages: English: Fluent (C1), German: Elementary (A2), Persian: Native

Selected Publications

- 1 [Shakibhamedan, Salar, Nima Amirafshar, Nima Taherinejad, and Axel Jantsch \(2025\)](#). "Heterogeneous Efficient Vision Models Bridging Accuracy and Efficiency (Under Revision -IEEE Transaction)". In.
- 2 [Nagesh, Nitish*, Shakibhamedan, Salar*, Mahdi Bagheri, Ziyu Wang, Nima TaheriNejad, Axel Jantsch, and Amir M Rahmani \(2025\)](#). "FairTabGen: Unifying Counterfactual and Causal Fairness in Synthetic Tabular Data Generation". In: *arXiv preprint arXiv:2508.11810*.
- 3 [Shakibhamedan, Salar, Nima Amirafshar, Ahmad Sedigh Baroughi, Hadi Shahriar Shahhoseini, and Nima Taherinejad \(2024\)](#). "ACE-CNN: Approximate Carry Disregard Multipliers for Energy-Efficient CNN-Based Image Classification". In: *IEEE Transactions on Circuits and Systems I: Regular Papers*.
- 4 [Salar Shakibhamedan, Anice Jahanjoo, Amin Aminifar, Nima Amirafshar, Nima TaheriNejad, and Axel Jantsch \(2024\)](#). "An Analytical Approach to Enhancing DNN Efficiency and Accuracy Using Approximate Multiplication". In: *ICML 2024 Workshop on Advancing Neural Network Training*.
- 5 [Dewnant Katore*, Salar Shakibhamedan*, Nima Amirafshar, Nima TaheriNejad, Axel Jantsch, Marijn Janssen, Aaron Yi Ding \(Submitted\)](#). "Approximation Strategies for Vision Models on Edge Devices: An Accuracy-Efficiency Trade-off". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence*.
- 6 [Shakibhamedan, Salar, Kooshan Hashemifard, Farhad Faradji, and Mansour Vali \(May 2016\)](#). "Persian Musical Instrument Recognition System". In: *International Conference on New Research Achievements in Electrical and Computer Engineering*.

Honors and Awards

2024 **Marshall Plan Scholarship**, Vienna

- Knowledge transfer program USA ↔ Austria (10% acceptance rate).

KUWI Research Grant, TU Wien, Vienna

2023 **DAC Young Fellowship Program**, San Francisco.

2019-Now **Certifications:** See LinkedIn profile for details.